Modeling the Economic Consequences of Nurse Staffing Decisions in Pandemic Hospital Systems: A COVID-19-Based Analysis

Yuke Hou (1913922), Xiangwen Gao (1892711), Anna Tam Ly (1915394), Saung Hnin Phyu (1931834), Samara Pires (1864538)

Contributions of each group member

Yuke Hou:

(1) Wrote and refined all numerical-simulation code, implementing the full compartment update for both waves. (2) Created and formatted all numerical-simulation figures, including external-input plots and staffing trajectories. (3) Interpreted the numerical-simulation results across redeployment scenarios and explained their operational implications.

Samara Pires:

(1) Conducted literature research to find the data for the province of Alberta (2) Performed comprehensive data cleaning, exploratory analyses, and data visualizations to identify the most appropriate time period for the study. (3) Development of the mathematical model and model equations for nurse workforce analysis. (4) Wrote numerical simulation code.

Xiangwen Gao:

(1) Performed local sensitivity testing to evaluate how changes in key variables affected overall model dynamics and projections. (2) Interpreted simulation results and validated them against epidemiological mechanisms. (3) Contributed to model discussion and modification.

Anna Tam Ly:

(1) Wrote the model assumptions; (2) Drew the model diagram; (3) Created the parameter table and found values from literature (except for nurse-to-patient ratio); (4) Created the model equations; (5) Calculated the initial workforce population; (6) Calculated the weekly infection risk; (7) Wrote the code for the nurse model function script

Saung Hnin Phyu:

(1) Conducted literature searches for background and context; (2) Cleaned and organized data to determine the appropriate project time period; (3) Completed the cost analysis, including data preparation and financial calculations; (4) Took part in reviewing and modifying model diagram

Abstract

Hospitals faced major staffing challenges during the COVID-19 pandemic as patient numbers surged and many nurses were absent due to illness. To understand how different staffing strategies affect both workforce capacity and hospital costs, we built a model that simulates nurse availability and demand during epidemic waves. The model tracks how many nurses are working, absent, or redeployed from other units, and calculates the costs of covering shortages through overtime or redeployment.

Using Alberta hospitalization data, we ran simulations for two COVID-19 waves (Alpha and Omicron) and tested different staffing policies. We found that the number of nurses needed to backfill positions rises sharply as infections increase, and the balance between overtime and redeployment meaningfully shapes overall costs and workforce distribution. Among the strategies examined, one mixed approach consistently offered a more stable and cost-efficient solution while limiting strain on non-COVID units. The sensitivity analysis showed that nurse availability is most affected by changes in key staffing parameters, especially the nurse-to-patient ratio, and that these effects intensify as the epidemic wave progresses.

Introduction

Background

Nurse staffing levels directly influence hospital performance, patient safety, and workforce sustainability. Adequate nurse-to-patient ratios are associated with improved outcomes such as reduced mortality, fewer adverse events, and shorter hospital stays [1]. In contrast, understaffing increases workload, accelerates burnout, and raises turnover rates, which diminishes the availability of experienced staff and compromises quality of care [2].

The COVID-19 pandemic exposed the fragility of hospital staffing systems. Surges in patient demand coincided with staff absences due to infection and quarantine, creating acute shortages. High workloads also increased the risk of hospital-acquired infections (HAIs), as nurses struggled to maintain monitoring and infection control protocols under pressure [3]. These clinical challenges were compounded by financial ones: reliance on overtime, agency nurses, and redeployment imposed substantial costs. In Alberta, for example, spending on agency and travel nurses reached approximately CAD 330 million over three years [4]. Such expenditures highlight the need for integrated approaches that consider both workforce resilience and economic sustainability.

Project Description

This project develops a health economic model to evaluate hospital staffing dynamics during epidemic conditions. The model integrates epidemiological processes (nurse infection and recovery), operational strategies (redeployment and overtime), and economic costs (wage differentials with three different staffing approaches). Using hospitalization data from Alberta, we implemented numerical simulations to capture weekly staffing flows across susceptible, infected, recovered, and redeployed nurses.

In addition to simulation, we conducted sensitivity analyses to examine how variations in nurse-to-patient ratios and redeployment proportions affect change in the system dynamics. A detailed cost analysis was performed to compare the financial burden of different staffing strategies, quantifying expenditures under varying redeployment and overtime scenarios. Together, these components provide an integrated assessment of how staffing policies influence hospital capacity and economic sustainability during epidemic surges.

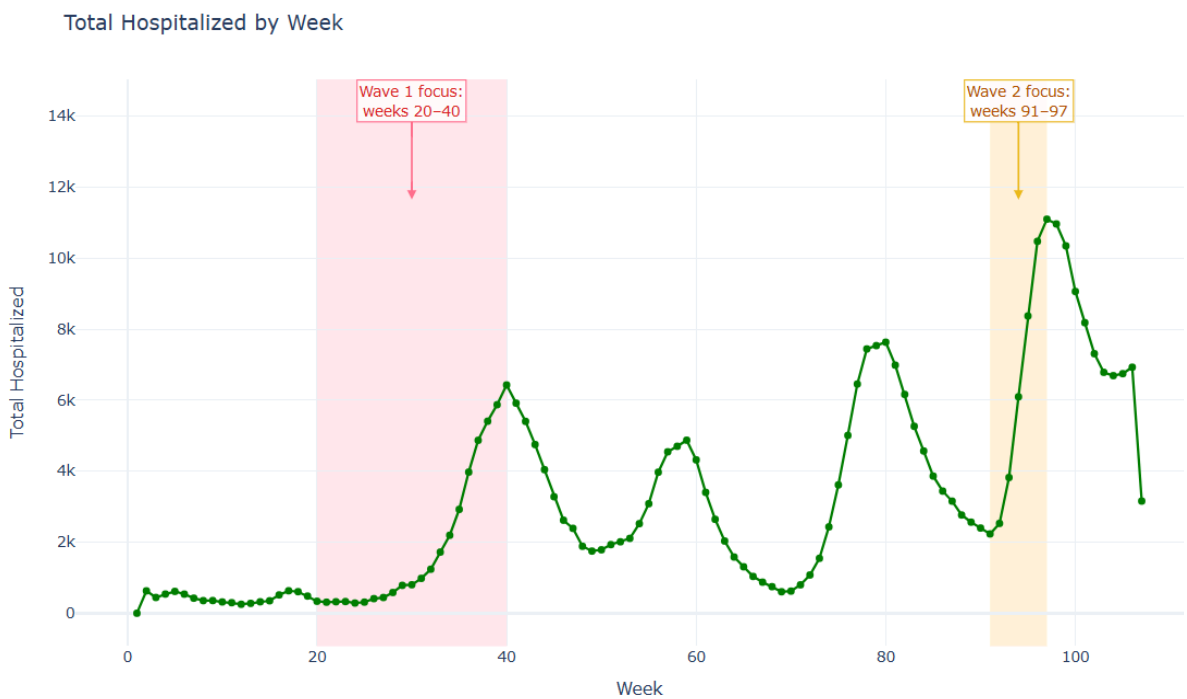
Research Objective

The objective of this study is to estimate the number of nurse positions that must be backfilled, expressed in full-time equivalents (FTEs), and to quantify the economic consequences of alternative staffing decisions under epidemic conditions.

Methods

Data

The COVID-19 data for our project was sourced from the Esri Canada COVID-19 Resources Hub's "Provincial Daily Totals" dataset [5], which provides daily net new case information for each Canadian province by date. Hosted on ArcGIS Hub, this resource compiles provincial epidemiological data for pandemic tracking and analysis. This dataset provided total hospitalizations by province from April 2020 to April 2022. We focused on Alberta's total hospitalization data, visualizing peaks and waves to select two periods: the Alpha (Aug 12 - Dec 30, 2020) and Omicron (Dec 28, 2021 - Feb 2, 2022): both capturing the upward curve to peak demand. These periods enabled analysis of maximum nurse requirements, contrasting the gradual 20-week Alpha increase with Omicron's rapid 7-week surge to quantify hospitalization impacts on nurse workload. The serology data used to estimate weekly community infection risk for the Alpha and Omicron waves was obtained from the COVID-19 Immunity Task Force website [6]. The data visualization is shown below:



Model

In order to achieve our objective of quantifying the economic consequences of different staffing decisions under epidemic conditions, we constructed a deterministic, discrete-time compartmental health workforce model using a system of difference equations.

Assumptions

Workforce:

The hospital has a fixed maximum number of active nurse positions, denoted A_{max}, and staffing is measured in full-time equivalents (FTEs), where one full-time 40-hour nurse corresponds to 1.0 FTE, a 20-hour part-time nurse corresponds to 0.5 FTE, and two part-time 0.5-FTE nurses together equal one full-time nurse. Nurses are assigned either to the COVID-unit (AC) or to other units (AO), and once a nurse from another unit is redeployed to the COVID-unit, they remain there for the entire duration of the wave. Within the COVID-unit, nurses are categorized as active susceptible (AS), infected and absent (I), or active recovered (AR), while nurses in other units are assumed to be fully susceptible because they have never been previously infected. The model does not include recruitment, training pipelines, or external inflows.

Infection:

We consider two separate COVID waves from the beginning until the peak as mentioned in the data section (Alpha and Omicron). COVID patients are kept in a separate unit from other patients and are only looked after by active nurses assigned to the COVID-unit. COVID-unit nurses are not getting infected inside the hospital, only externally via community infections (where new infections are proportional to the weekly community infection risk and the susceptible nurses at that week) and once infected, nurses are quarantined and cannot infect other susceptible nurses. For the weekly community infection risk, we assumed that the increase in weekly community seroprevalence is linear. We do not consider reinfection within each wave or other public health interventions such as vaccination.

Demand:

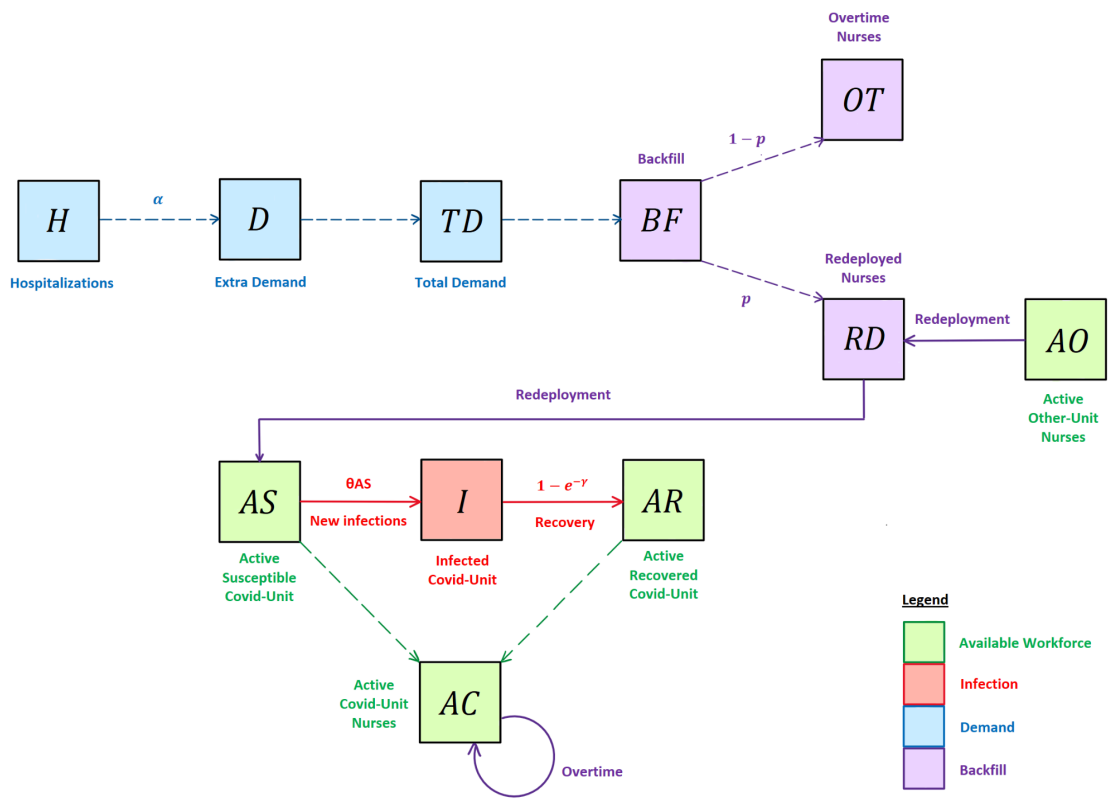
The baseline nurse demand is set equal to the initial number of active susceptible nurses in the COVID-unit, and additional nurse demand is proportional to the number of COVID hospitalizations and the nurse-to-patient ratio.

Backfill:

Backfill FTEs are covered by overtime (from COVID-unit only) and straight-time redeployment (from other units), with wage rates C_{OT}, C_{ST} where overtime pay is 1.5 times the regular wage rate. We cap the number of redeployed nurses, and any remaining staffing needs are covered by overtime nurses.

<u>Diagram</u>

The below mathematical model illustrates how the ***available workforce*** (**green**) responds to reduced capacity due to ***infection***-related absences (**red**) and increased nurse ***demand*** due to COVID hospitalizations (**blue**), as well as how staffing shortfalls are met through ***backfill*** (**purple**).



Description of Parameters

Category	Parameter	Description	Value	Source
Workforce	AS_0	Initial susceptible COVID-unit nurses	6776	[7], [8], (Appendix Page 1)
Workforce	AO_0	Initial other-unit nurses	19,287	[7], [8], (Appendix Page 1)
Infection	m	Weekly increase in community seroprevalence	2.0783×10^{-5} (Alpha); 2.1188×10^{-4} (Omicron)	[9], (Appendix Table 1)
Infection	d	Average absence duration for infection	2 Unit: week^{-1}	[10]
Infection	γ	Recovery rate	$\frac{1}{d} = \frac{1}{2}$ Unit: week^{-1}	[10]
Demand	α	Nurse-to-patient ratio	1: 5 (1 nurse to 5 patients)	[11]
Backfill	C_{ST}	Cost per hour of redeployment	47.38 Unit: hour^{-1}	[12]

Backfill	C_{OT}	Cost per hour of overtime	$47.38 \times 1.5 = 71.07$ Unit: hour^{-1}	[12], [13]
Backfill	$p; (1 - p)$	Proportion of redeployed FTE nurses; proportion of overtime FTE nurses	0%, 50%, 100%	Assumed
Backfill	W	Hours per FTE	40 Unit: week^{-1}	Assumed

Description of Variables and Equations

Variable/Equation	Description
H_t	Number of COVID-19 hospitalizations
$D_t = \alpha \cdot H_t$	Extra demand for nurses based on hospitalizations
$TD_t = AS_0 + D_t$	Total demand of nurses
$BF_t = \max\{0, TD_t - AC_t\}$	Backfill demand (nurses needed to be filled)
$RD_t = \min\{AO_t, p \times BF_t\}$	Number of nurses redeployed from other units
$OT_t = (1 - p) \times BF_t$	Number of overtime nurses in COVID-unit
$A_{max} = AS_t + I_t + AR_t + AO_t, \forall t$	Maximum number of available nurse positions in the hospital
$AC_t = AS_t + AR_t$	Active COVID-unit nurses
$AO_{t+1} = AO_t - RD_t$	Active other-unit nurses
$\theta_t = \int_t^{t+1} mx\, dx$	Infection risk of COVID-19 infection among nurses in the COVID-unit (Appendix Table 1)
$\text{new_inf}_t = \min\{AS_t, \theta_t \times AS_t\}$	Number of new nurses infected/absent
$\text{return}_t = (1 - e^{-\gamma \Delta t}) \times I_t$	Number of nurses returning after infection
$AS_{t+1} = AS_t - \text{new_inf}_t + RD_t$	Susceptible active COVID-unit nurses
$I_{t+1} = I_t + \text{new_inf}_t - \text{return}_t$	Infected inactive COVID-unit nurses
$AR_{t+1} = AR_t + \text{return}_t$	Recovered active COVID-unit nurses
$\text{cost}_t = (C_{OT} \times W \times RD_t) + (C_{ST} \times W \times OT_t)$	Cost per week of redeployed and overtime nurses

Numerical Simulation

The simulations use the component structure and the weekly update equations from the Model section. We simulate nurse staffing during the Alpha and Omicron waves with a time step Δt = 1 week. All parameter values come from the earlier parameter table, including the fixed workforce size $Amax$ and the recovery rate γ.

The simulations use three external inputs: the weekly hospitalization counts H_t, the resulting extra staffing demand D_t, and the infection-risk sequence θ_t. These inputs determine the conditions for each wave each week and do not affect any part of the updated model equations.

At each weekly step, the model updates staffing flows based on the rules defined in the Model section. The backfill need arises from the gap between required and available COVID-unit nurses. This need is allocated to overtime and redeployment according to the chosen proportion p. Redeployment is limited by the number of nurses in the other unit and by the safety rule that no more than 50% of the initial other-unit workforce can be transferred. Across each wave, the total number of active nurses remain constant ($AS_t + I_t + AR_t + AO_t = AC_t + I_t + AO_t = Amax, \forall t$).

The simulations generate weekly trajectories of AS, I, AR, AC and AO for each wave and redeployment level. The corresponding demand curves are recorded as well. These outputs form the basis for the analysis in the Results section.

Local Sensitivity Analysis

To assess how the change in the main parameters will affect system dynamics, especially the flow in each "stock", we also analyzed local sensitivity of each parameter. We perturbed each parameter by $\pm$ 1% while maintaining the other one constant and recorded the proportional changes across all nurse-related compartments.

As the sensitivity index of the parameters varies across time, we also incorporated this dynamics in our analysis.

Cost Analysis

Parameters and Units

To evaluate the economic implications of different staffing strategies, we first establish the cost parameters associated with full-time equivalents (FTEs), overtime, and redeployment.

Full-time equivalent (FTE)	40 working hrs/week
Base Hourly Wage	47.38 CAD/hr
Overtime	20 hrs/week (0.5 FTE) at 1.5 × base wage = 71.07 CAD/hr
Redeployment	40 hrs/week (0.5 FTE) at base wage = 47.38 CAD/hr
Weekly Cost per FTE for Overtime	1,421.4 CAD per 0.5 FTE
Weekly Cost per FTE for Redeployment	1,895.2 CAD per 1 FTE

Staffing scenarios are defined by the proportion of backfill covered by redeployment, denoted $p \in \{0, 0.5, 1\}$. When $p = 0$, all backfill is covered by overtime; when $p = 1$, all backfill is covered by redeployment.

Cost Equations

Let I_t represent the number of absent FTE nurses in week t. Weekly costs are calculated as:

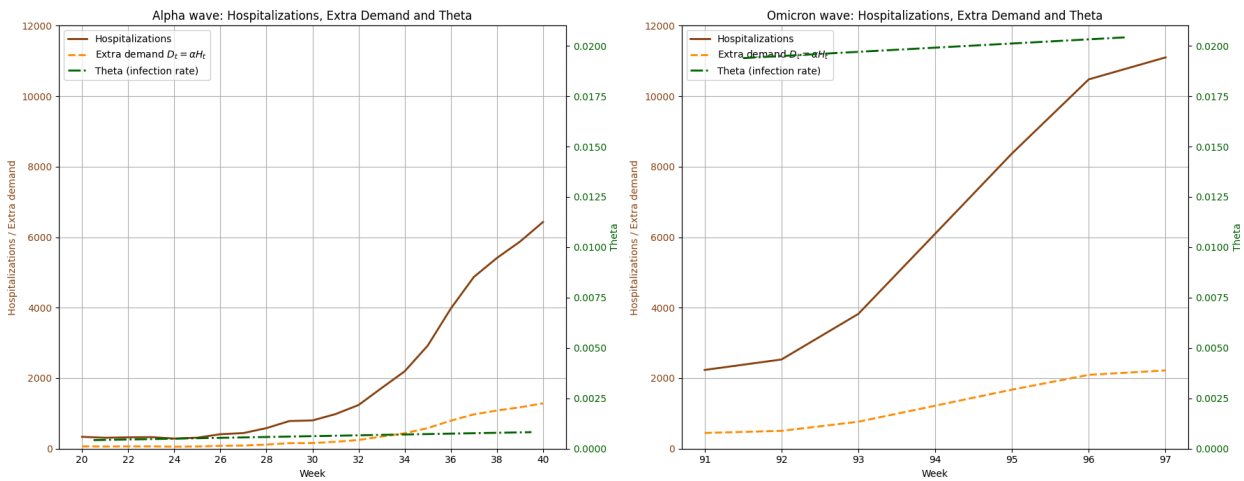
$$Cost\ (t,\ p)\ = C_{ST,w}\ \cdot p\ \cdot\ I_t\ +\ C_{OT,w}\ \cdot 2\ \cdot\ (1\ -\ p)\ \cdot\ I_t$$

Thus:

- For $p\ =\ 0$: $Cost\ (t,\ 0)\ = 2\ \cdot\ I_t \cdot\ C_{OT,w}$
- For $p\ =\ 0.5$: $Cost\ (t,\ 0.5)\ = I_t\ (\frac{1}{2} \cdot\ C_{ST,w}\ +\ C_{OT,w})$
- For $p\ =\ 1$: $Cost\ (t,\ 1)\ = I_t \cdot C_{ST,w}$

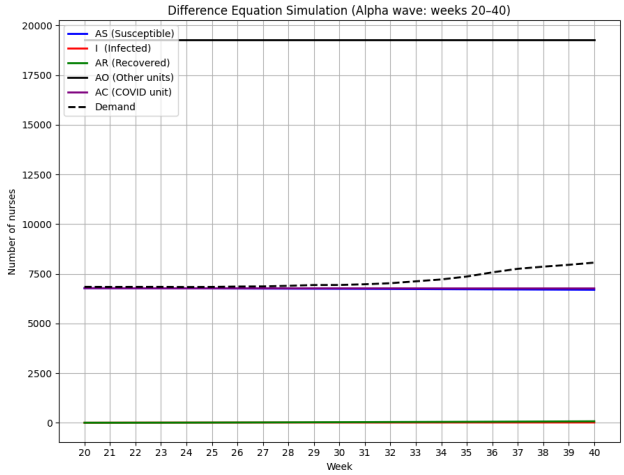
Results

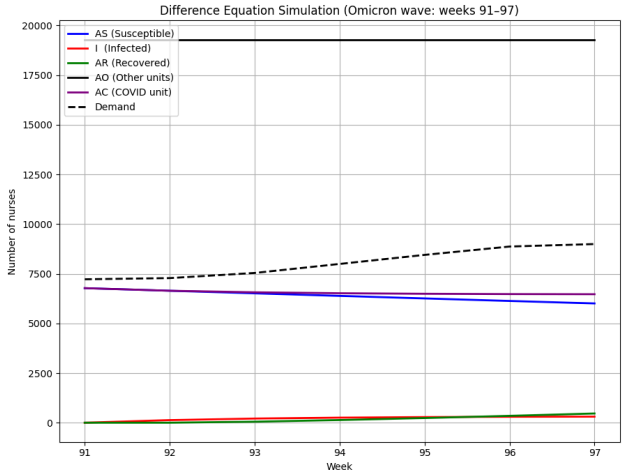
<u>Numerical Simulation</u>



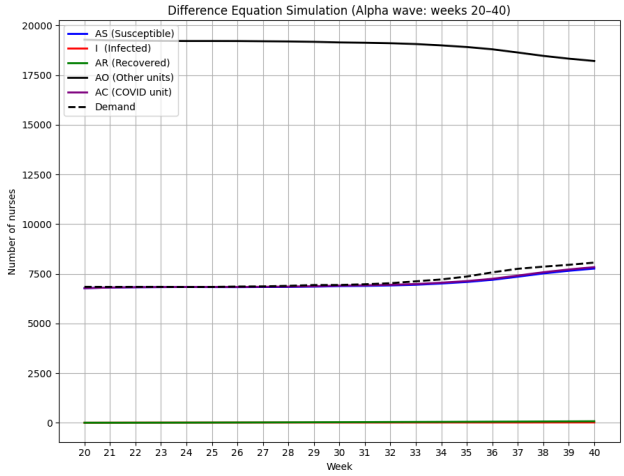
During the Alpha wave, hospitalizations H_t rise gradually from about 300 to 6500 cases per week. The extra staffing demand D_t and the infection risk θ_t follow the same gradual and low-level upward pattern. The Omicron wave shows a much sharper increase. H_t exceed 11000 within only 7 weeks, and both D_t and θ_t reach levels several times higher than in the Alpha wave. With shared y-axis limits, the contrast becomes clear: Omicron presents a far more intense and rapidly increasing shock to the nursing system.

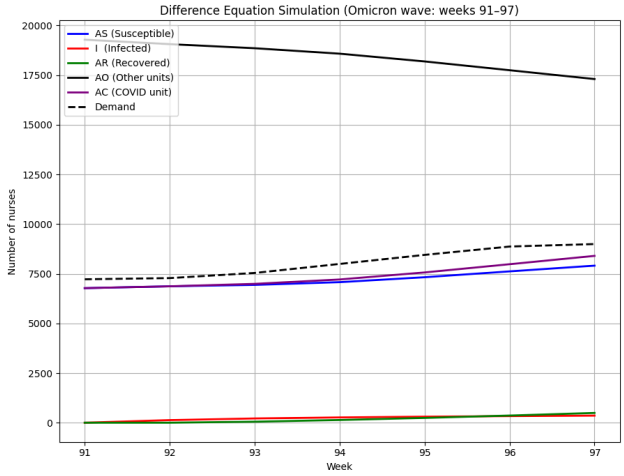
0% Redeployment



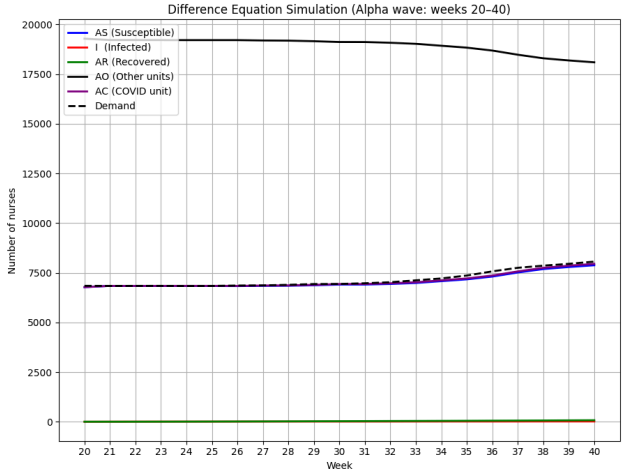


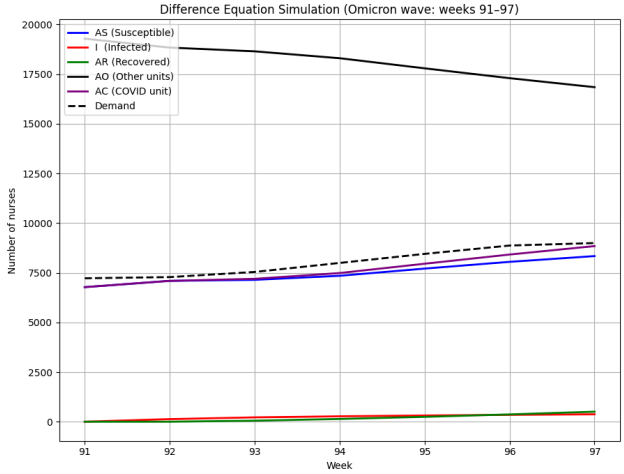
50% redeployment





100% redeployment





At 0% redeployment, all backfill is covered by overtime. During the Alpha wave, low infections keep I_t and AC_t nearly stable because recoveries offset new infections. AC_t initially meets demand but later falls below it as hospitalizations rise. In the Omicron wave, the higher infection risk leads to rapid growth in I_t and a sharper decline in AC_t, resulting in severe understaffing.

At 50% redeployment, transfers supply half of the backfill and expand the susceptible pool. As a result, AC_t begins to rise rather than flatten or decline. In the Alpha wave, COVID-unit staffing increases steadily under low infection pressure. In the Omicron wave, redeployment offsets much of the loss caused by rising infections and keep AC_t closer to demand. The shift from 0% to 50% produces the largest improvement in reducing the staffing gap.

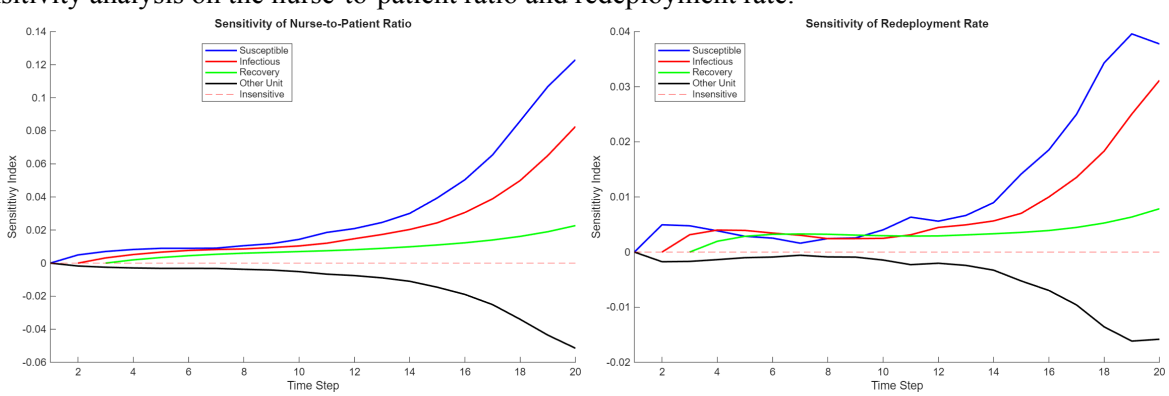
At 100% redeployment, all backfill comes from transfers. In both waves, AC_t rises more strongly than under the 50% scenario. The increase in AC_t does not fully match the drop in AO_t because some redeployed nurses become infected. The gain from raising redeployment from 50% to 100% is smaller than the earlier increase from 0% to 50%. Most of the staffing gap is already reduced once half of the backfill is covered by redeployment (See Appendix Figures 1–3 for plots separated by compartment).

<u>Local Sensitivity Analysis</u>

The parameter baseline is set as follows:

Parameter	**Baseline**
Nurse-to-patient ratio (α)	1:5
Redeployment rate (p)	50%

Focusing on the Alpha Wave (mapped to Time Steps 0–20 in the graphs), we performed local sensitivity analysis on the nurse-to-patient ratio and redeployment rate.



The time plots reveal that both the nurse-to-patient ratio and redeployment rate exert a positive influence on all COVID unit compartments. The hierarchy of the sensitivity plots is similar: highest for the Susceptible compartment, followed by the Infectious and then the Recovered compartments. Furthermore, both sensitivity indices amplify over time, particularly as the wave peaks, indicating that nurse demand in the COVID units becomes increasingly sensitive to these parameters as the infection spreads. Despite the similarity there are some minor differences between the plots. The first one lies in the magnitude. The redeployment rate is approximately three times less influential over the span of 20 weeks, peaking at a sensitivity index of roughly 0.04 compared to 0.12 for the nurse-to-patient ratio. Another

notable distinction occurs in the first half of the simulation (time step 0–10), where the impact of redeployment rate on COVID unit compartments exhibits minor fluctuations. This volatility likely reflects the initial dynamics of the wave: the demand for nurses in the COVID unit rises gradually at the onset, meaning the redeployed workforce is initially sufficient to meet requirements.

Conversely, the impact of both parameters on the other unit is negative. Notably, the magnitude of the negative sensitivity is approximately half that of the Susceptible compartment, which is consistent with the model's predefined 50% redeployment constraint.

In conclusion, the local sensitivity analysis quantifies the temporal evolution and magnitude of parameter influence on nurse-related compartments from the wave's onset to its peak. These indices serve as a strategic guide for intervention: to effectively manage staffing levels at any specific point in time, decision-makers should prioritize adjusting parameters with the highest sensitivity indices, as these offer the greatest leverage for controlling the system.

Cost Analysis

During Alpha, the number of infected nurses increased gradually from 2.9 to 15 FTEs. Weekly costs rose accordingly, with total expenditures over the 20-week period amounting to:

- 0% redeployment (all overtime): 567,171 CAD
- 50% redeployment: 378,114 CAD and
- 100% redeployment: 330,850 CAD

The results demonstrate that redeployment substantially reduces costs compared to reliance on overtime alone.

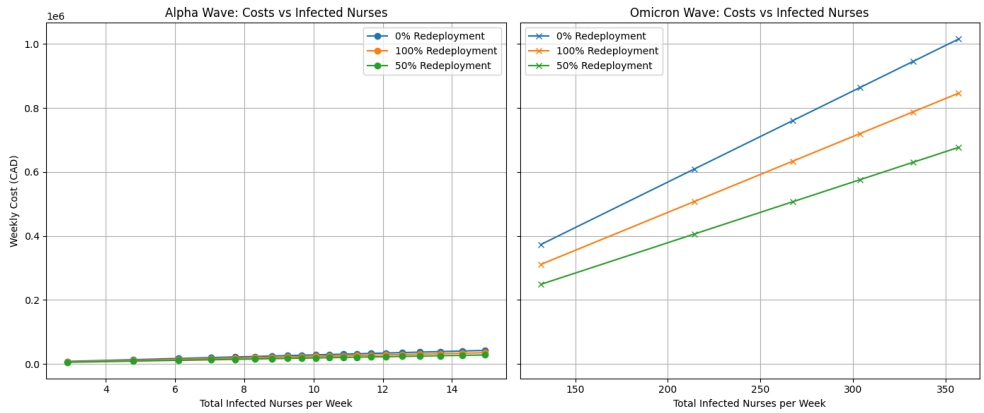
During Omicron, the number of infected nurses increased sharply, rising from 131 to 357 full-time equivalents (FTEs). This escalation led to a substantial rise in weekly staffing costs, with total expenditures reaching

- 0% redeployment (all overtime): 4,566,855 CAD
- 50% redeployment: 3,044,570 CAD and
- 100% redeployment: 2,663,999 CAD

These results demonstrate the significant financial strain hospitals face during severe epidemic surges. While redeployment strategies consistently reduced costs compared to reliance on overtime alone, they also required shifting staff away from non-COVID units, which may limit capacity in other areas of hospital operations.

The detailed weekly cost tables for both waves which show infected nurse counts and total weekly expenditures under each staffing scenario, are provided in the Appendix (Table 2,3) for reference. These tables enable direct comparison across the 0%, 50%, and 100% redeployment strategies and illustrate how costs evolve as weekly infections increase.

The following two figures illustrate the weekly staffing costs for Wave 1 and Wave 2 under different redeployment strategies. Costs increase with the number of infected nurses, with overtime-only coverage resulting in the highest expenses, full redeployment the lowest, and a 50% mixed strategy providing an intermediate, more balanced cost outcome.



Weekly Overtime Burden: Number of Nurses Needed

Overtime limits strongly affect how many nurses are needed to cover absences. With a 4-hour cap, 10 nurses are required to replace one full-time equivalent (FTE), with an 8-hour cap, 5 nurses are needed and with a 20-hour cap, only 2 nurses are sufficient.

For 0% Redeployment and 100 % Overtime:

Total Overtime Hours	$I_t \times 40$ hours
Total nurses required for 4 hours overtime	$I_t \times 10$ nurses
Total nurses required for 8 hours overtime	$I_t \times 5$ nurses
Total nurses required for 20 hours overtime	$I_t \times 2$ nurses

During Alpha Wave, the number of infected nurses increased gradually from fewer than 3 to about 11 FTEs. Depending on the overtime cap, between 29 and 150 nurses were required to share the load at 4 hours/week, compared with 6 to 30 nurses at 20 hours/week. In the Omicron Wave, infection levels rose much more sharply, from 131 to 276 FTEs. Overtime requirements escalated to over 14,000 hours per week. At the strictest cap (4 hours/week), more than 3,500 nurses were needed to absorb the overtime, while at 20 hours/week the requirement dropped to fewer than 750 nurses.

For 50% Redeployment and 50 % Overtime:

In this scenario, half of the required nurses will be backfilled from the nurses as FTE from another unit.

Total Overtime Hours	$I_t \times 20$ hours
Total nurses required for 4 hours overtime	$I_t \times 5$ nurses
Total nurses required for 8 hours overtime	$I_t \times 2.5$ nurses
Total nurses required for 20 hours overtime	$I_t \times 1$ nurse

A 50% redeployment scenario reduced the overtime burden by half, since part of the backfill was covered by full-time redeployed nurses. Even so, the scaling effect remained as lower overtime caps required substantially more nurses to share the workload, while higher caps concentrated the burden on fewer individuals.

Detailed weekly values for both waves and scenarios are provided in the Appendix (Tables 4-7). These results underscore that while redeployment reduces financial costs and overtime hours, the choice of overtime cap strongly affects workforce distribution and the risk of burnout.

Discussion

The results of this study show that staffing strategies during epidemic waves strongly affect both workforce capacity and hospital costs. The Alpha wave increased slowly, allowing recoveries to balance new infections and keeping shortages relatively small. In contrast, the Omicron wave rose quickly, producing much higher infection levels and costs. This difference shows how the speed and intensity of epidemic waves directly change the feasibility of staffing responses. Similar patterns have been observed in Canadian hospitals, where sudden surges in patient demand created acute staffing challenges [2].

Redeployment consistently lowered costs compared to overtime alone. The largest improvement occurred when moving from 0% to 50% redeployment, while the extra benefit of increasing redeployment to 100% was smaller. Sensitivity analysis also showed that the nurse-to-patient ratio had the strongest effect on staffing needs, with redeployment rate playing a smaller role. This supports evidence that safe nurse-to-patient ratios are linked to better patient outcomes, including reduced mortality and shorter hospital stays [1]. It also agrees with research showing that understaffing increases the risk of hospital-acquired infections [3].

In the Alpha wave, the main issue was cost rather than feasibility. With a 20-hour overtime cap, all three strategies, which are 0% redeployment, 50% redeployment, and 100% redeployment, were possible. The maximum number of nurses needed to share overtime duties never exceeded 30, which is manageable. Therefore, in Alpha the choice of strategy depended mainly on cost, with redeployment reducing expenses compared to overtime alone. In the Omicron wave, the situation was different. Relying only on overtime was not possible because the number of absent nurses was too high, even under the 20-hour cap. Although 100% redeployment was the cheapest option, it required moving a very large number of nurses from non-COVID units, which would weaken hospital capacity in other areas. This shows that affordability alone cannot determine the best solution.

The mixed 50% redeployment strategy gave the most balanced outcome during Omicron. It lowered costs compared to overtime-only coverage, while avoiding the excessive transfers required under full redeployment. This approach kept COVID-unit staffing closer to demand without severely reducing non-COVID services. For public health decisionmakers, these findings are important because they show that mixed strategies are more sustainable in fast, high-pressure waves, while all options remain usable in slower waves. This evidence can guide hospital leaders in balancing cost efficiency, workforce sustainability, and patient safety during epidemic surges.

References

[1] Dall'Ora C., Griffiths P., Ball J., Simon M., Aiken L.H. Association of nurse staffing with patient outcomes: a systematic review. BMJ Open, 2022. Available from: https://pubmed.ncbi.nlm.nih.gov/35780608

[2] Bae S.H., Mark B.A. Nurse staffing, turnover, and patient outcomes: evidence from Canadian hospitals. Health Care Management Science, 2022. Available from: https://pubmed.ncbi.nlm.nih.gov/29685321/

[3] Shang J., Needleman J., Liu J., Larson E. Nurse staffing and healthcare-associated infections: unit-level analysis. BMJ Quality & Safety, 2019. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC6478399/

[4] The Progress Report. Alberta Health Services to spend $330 million on agency nurses since 2022. 2024. Available from: https://www.theprogressreport.ca/exclusive_ahs_projects_to_spend_330_million_on_agency_nurses_since_2022

[5] COVID-19 Hospitalisation Data for Alberta [Internet]. Resources COVID-19 Canada; [cited 2025 Dec 10]. Available from: https://resources-covid19canada.hub.arcgis.com/datasets/provincial-daily-totals/explore?showTable=true

[6] Serology (Infection Rate) Data [Internet]. COVID-19 Immunity Task Force; [cited 2025 Dec 10]. Available from: https://www.covid19immunitytaskforce.ca/seroprevalence-in-canada/

[7] Alberta Sector Profile: Health Care - Job Bank [Internet]. Jobbank.gc.ca. 2025 [cited 2025 Dec 10]. Available from: https://www.bc.jobbank.gc.ca/trend-analysis/job-market-reports/alberta/sectoral-profile-health-care

[8] Mensinger JL, Brom H, Havens DS, Costello A, D'Annunzio C, Durning JD, et al. Psychological responses of hospital-based nurses working during the COVID-19 pandemic in the United States: A cross-sectional study. Applied Nursing Research [Internet]. 2022 Feb 1;63:151517. Available from: https://www.sciencedirect.com/science/article/pii/S0897189721001245

[9] Caminsky N. Seroprevalence in Canada [Internet]. COVID-19 Immunity Task Force. Available from: https://www.covid19immunitytaskforce.ca/seroprevalence-in-canada/

[10] Alberta Government [Internet]. Self-isolation information sheet. 2020 [cited 2025 Dec 10]. Available from: https://open.alberta.ca/publications/self-isolation-information-sheet/resource/91cd86ac-7eaf-4376-9f9a-0b600a9af2fa

[11] Ernest and Young (EY) [Internet]. Alberta Health Services Performance Review Final Report. 2019. Available from: https://www.alberta.ca/alberta-health-services-review

[12] ALIS [Internet]. Registered Nurse. 2018. Available from: https://alis.alberta.ca/occinfo/occupations-in-alberta/occupation-profiles/registered-nurse/

[13] Government of Alberta [Internet]. Employment Standards Rules – Overtime Hours and Overtime Pay. Available from: https://www.alberta.ca/overtime-hours-overtime-pay

Appendix

Dataset:

- Hospitalization Data for Nurse Staffing Analysis

Code for various modules:

- Data Cleaning and Visualisation
- Numerical Simulation
- Sensitivity analysis; nurse model
- Cost Analysis

A copy of proposal

https://docs.google.com/document/d/1cTTMZbsLYO4WdFpe86qCKPmAnxXJJXeBu5u9WBUBEJg/edit?usp=sharing

Initial nurse workforce calculation

Number of nurses in Alberta: 38,900

Percentage of nurses working in hospitals: 67%

Number of nurses working in hospitals in Alberta: $38,900 \times 0.67 = 26,063$

Percentage of nurses working in COVID-unit: 26%

Initial number of nurses working in COVID-unit in Alberta: $26,063 \times 0.26 = 6776$

Initial number of nurses working in other-unit in Alberta: $26,063 - 6776 = 19,287$

Table 1: Weekly community infection risk calculation

	Alpha	**Omicron**
Community seroprevalence from data	August 2020: 0.00749 December 2020: 0.01996 Difference: 0.01247	December 2021: 0.1294 February 2022: 0.2489 Difference: 0.1195
Weekly increase in community seroprevalence	$\int_{20}^{40} mx\,dx = 0.01247$ $m = 2.0783 \times 10^{-5}$	$\int_{91}^{97} mx\,dx = 0.1195$ $m = 2.1188 \times 10^{-4}$
Weekly community infection risk	$\theta_{20} = \int_{20}^{21} mx\,dx = 0.0004260515$ $\theta_{21} = \int_{21}^{22} mx\,dx = 0.0004468345$ $\theta_{22} = \int_{22}^{23} mx\,dx = 0.0004676175$ $\theta_{23} = \int_{23}^{24} mx\,dx = 0.0004884005$ $\theta_{24} = \int_{24}^{25} mx\,dx = 0.0005091835$ $\theta_{25} = \int_{25}^{26} mx\,dx = 0.0005299665$	$\theta_{91} = \int_{91}^{92} mx\,dx = 0.01938822$ $\theta_{92} = \int_{92}^{93} mx\,dx = 0.01959590$ $\theta_{93} = \int_{93}^{94} mx\,dx = 0.01980338$ $\theta_{94} = \int_{94}^{95} mx\,dx = 0.02001006$ $\theta_{95} = \int_{95}^{96} mx\,dx = 0.02021674$ $\theta_{96} = \int_{96}^{97} mx\,dx = 0.02042342$

	$\theta_{26} = \int_{26}^{27} mx\,dx = 0.0005507495$ $\theta_{27} = \int_{27}^{28} mx\,dx = 0.0005715325$ $\theta_{28} = \int_{28}^{29} mx\,dx = 0.0005923155$ $\theta_{29} = \int_{29}^{30} mx\,dx = 0.0006130985$ $\theta_{30} = \int_{30}^{31} mx\,dx = 0.0006338815$ $\theta_{31} = \int_{31}^{32} mx\,dx = 0.0006546645$ $\theta_{32} = \int_{32}^{33} mx\,dx = 0.0006754475$ $\theta_{33} = \int_{33}^{34} mx\,dx = 0.0006962305$ $\theta_{34} = \int_{34}^{35} mx\,dx = 0.0007170135$ $\theta_{35} = \int_{35}^{36} mx\,dx = 0.0007377965$ $\theta_{36} = \int_{36}^{37} mx\,dx = 0.0007585795$ $\theta_{37} = \int_{37}^{38} mx\,dx = 0.0007793625$ $\theta_{38} = \int_{38}^{39} mx\,dx = 0.0008001455$ $\theta_{39} = \int_{39}^{40} mx\,dx = 0.0008209285$	

Table 2: Weekly Costs for Alpha (Aug - Dec 2020)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
20-21	2.8869	8206.879	6839.066	5471.253
21-22	4.7925	13624.12	11353.43	9082.746
22-23	6.0957	17328.86	14440.71	11552.57
23-24	7.0312	19988.3	16656.91	13325.53
24-25	7.7419	22008.67	18340.56	14672.45

25-26	8.3131	23632.48	19693.73	15754.99
26-27	8.8007	25018.63	20848.86	16679.09
27-28	9.2426	26274.86	21895.72	17516.58
28-29	9.6556	27448.94	22874.12	18299.29
29-30	10.0574	28591.18	23825.98	19060.78
30-31	10.4596	29734.55	24778.79	19823.03
31-32	10.8546	30857.46	25714.55	20571.64
32-33	11.2523	31988.04	26656.7	21325.36
33-34	11.6613	33150.74	27625.62	22100.5
34-35	12.0990	34395.04	28662.53	22930.02
35-36	12.5667	35724.61	29770.51	23816.41
36-37	13.0801	37184.11	30986.76	24789.41
37-38	13.6632	38841.74	32368.12	25894.5
38-39	14.3019	40657.44	33881.2	27104.96
39-40	14.9551	42514.36	35428.63	28342.91
Total Cost		567171	472642.5	378114

Table 3: Weekly Costs for Omicron (Dec 2021 – Feb 2022)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
91-92	131.3746	373471.7	311226.4	248981.1
92-93	214.2678	609120.5	507600.4	406080.3
93-94	267.4039	760175.8	633479.8	506783.9
94-95	303.8044	863655.1	719712.6	575770.1
95-96	332.3866	944908.6	787423.9	629939.1
96-97	357.2265	1015523	846269.6	677015.7
Total Cost		4566855	3805713	3044570

Table 4: Weekly Infected and Backfill Nurses (0% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	115	29	14	6
21-22	5.91	192	48	24	10
22-23	6.2	244	61	30	12
23-24	6.48	281	70	35	14
24-25	6.76	310	77	39	15
25-26	7.04	333	83	42	17
26-27	7.32	352	88	44	18
27-28	7.6	370	92	46	18
28-29	7.89	386	97	48	19
29-30	8.17	402	101	50	20
30-31	8.45	418	105	52	21
31-32	8.73	434	109	54	22
32-33	9.01	450	113	56	23
33-34	9.29	466	117	58	23
34-35	9.58	484	121	60	24
35-36	9.86	503	126	63	25
36-37	10.14	523	131	65	26
37-38	10.42	547	137	68	27
38-39	10.7	572	143	72	29
39-40	10.98	598	150	75	30

Table 5: Weekly Infected and Backfill Nurses (0% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	5255	1314	657	263
92-93	264.17	8571	2143	1071	429
93-94	267.04	10696	2674	1337	535
94-95	269.91	12152	3038	1519	608
95-96	272.78	13295	3324	1662	665
96-97	275.65	14289	3572	1786	714

Table 6: Weekly Infected and Backfill Nurses (50% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	58	14	7	3
21-22	5.91	96	24	12	5
22-23	6.2	122	30	15	6
23-24	6.48	141	35	18	7
24-25	6.76	155	39	19	8
25-26	7.04	166	42	21	8
26-27	7.32	176	44	22	9
27-28	7.6	185	46	23	9
28-29	7.89	193	48	24	10
29-30	8.17	201	50	25	10
30-31	8.45	209	52	26	10
31-32	8.73	217	54	27	11
32-33	9.01	225	56	28	11
33-34	9.29	233	58	29	12
34-35	9.58	242	60	30	12
35-36	9.86	251	63	31	13
36-37	10.14	262	65	33	13
37-38	10.42	273	68	34	14
38-39	10.7	286	72	36	14
39-40	10.98	299	75	37	15

Table 7: Weekly Infected and Backfill Nurses (50% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	2627	657	328	131
92-93	264.17	4285	1071	536	214
93-94	267.04	5348	1337	669	267
94-95	269.91	6076	1519	760	304
95-96	272.78	6648	1662	831	332
96-97	275.65	7145	1786	893	357

Figure 1: 0% Redeployment (subplot)

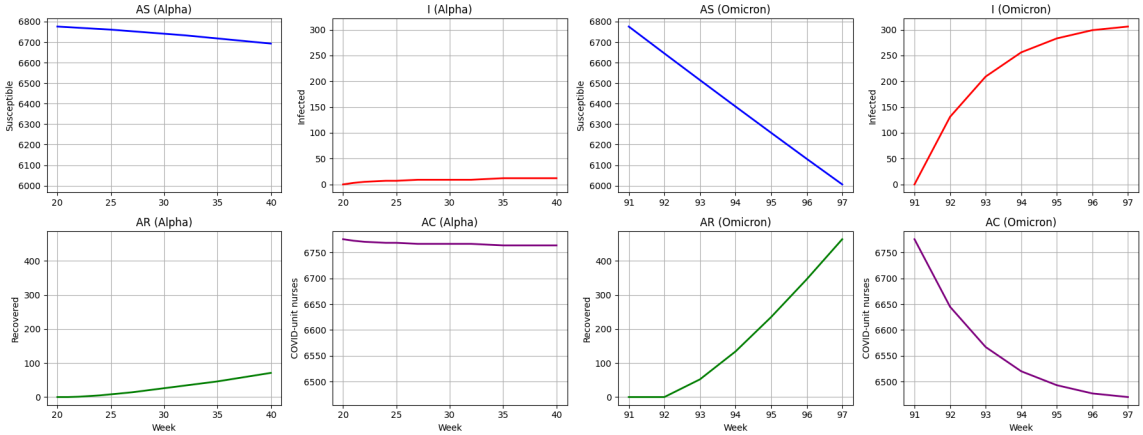


Figure 2: 50% Redeployment (subplot)

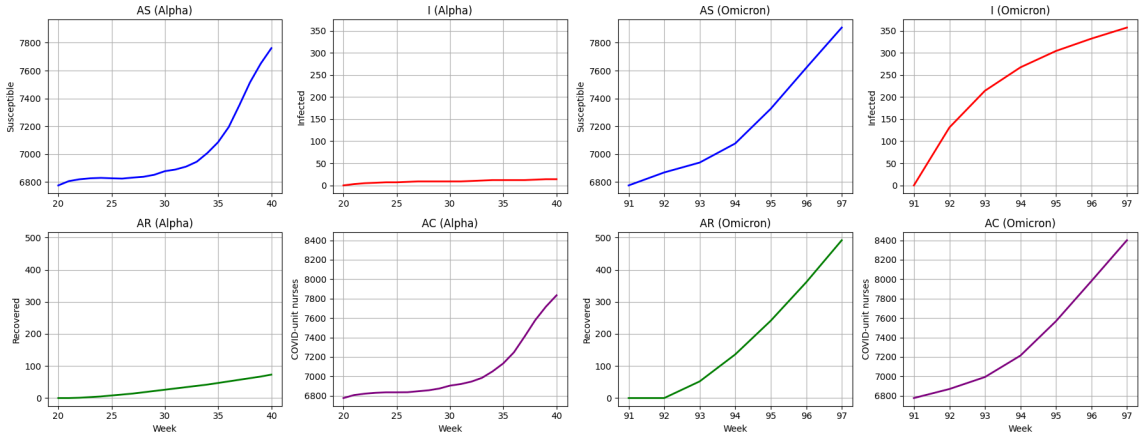
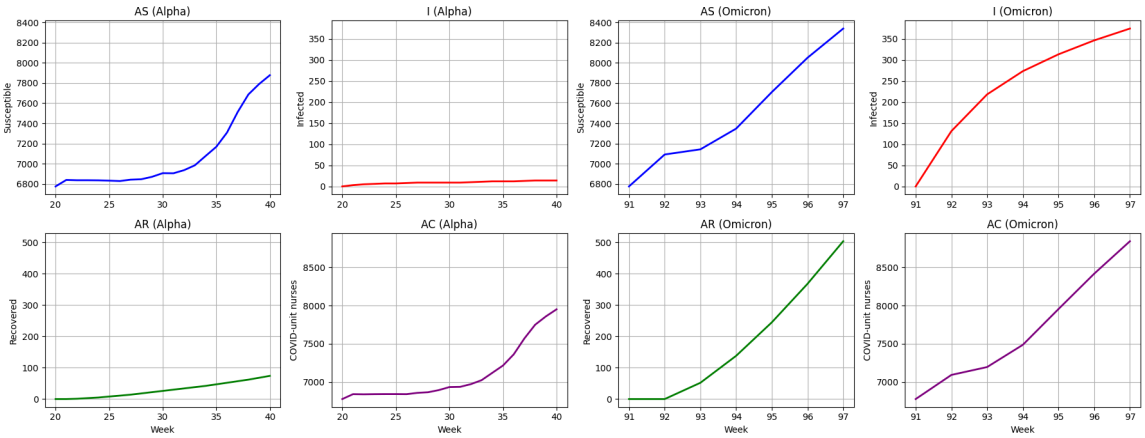


Figure 3: 100% Redeployment (subplot)



Briefing Note

Title: Advice to Alberta Health Services on Nurse Staffing Strategies During Epidemic Waves

Issue: Hospitals in Alberta faced severe nurse shortages during COVID-19 waves due to rising patient demand and staff absences. Decision-makers require clear information on how different staffing choices affect hospital costs and operations.

Purpose: To provide information on how redeployment and overtime strategies affect workforce capacity and hospital costs during different epidemic periods.

Background: Nurse staffing levels are directly linked to patient safety and hospital performance.
During COVID-19, rising hospitalizations happened at the same time as nurse absences, which led to serious shortages. Alberta spent approximately CAD 330 million on agency nurses over three years, highlighting the financial strain of emergency staffing. This project developed a compartmental health workforce model using Alberta hospitalization data for the Alpha (August - December 2020) and Omicron (December 2021– February 2022) waves.

Analysis: Numerical simulation results showed different patterns in the Alpha and Omicron waves. In the Alpha wave, shortages developed slowly. In the Omicron wave, shortages grew much faster, and covering them with overtime only was not possible. Redeployment helped reduce costs but required moving many nurses from non-COVID units, which created additional challenges. Sensitivity analysis shows that nurse-to-patient ratio was the most influential parameter, driving shortages more strongly than redeployment rate. Redeployment had a secondary effect but helped offset losses.
In the cost analysis, redeployment consistently lowered costs compared to overtime. 100% redeployment was cheapest but weakened non-COVID services. A 50-50 mix balanced affordability and service capacity, especially during Omicron.

Stakeholder Reaction: Nurses may resist excessive overtime due to burnout risk. Non-COVID units may oppose full redeployment, as it reduces their capacity. Patients and families expect safe staffing ratios across all hospital units.

Communications/ Key Messages:

- Epidemic surges require flexible staffing strategies.
- Redeployment reduces costs but must be balanced to protect non-COVID services.
- A mixed 50-50 redeployment and overtime approach is the most sustainable option in fast, high-pressure waves.

Financial Implications: During Alpha wave, costs ranged from CAD 567,171 (overtime only) to CAD 330,850 (full redeployment). In the Omicron wave, costs ranged from CAD 4.57 million (overtime only) to CAD 2.66 million (full redeployment). Mixed strategies reduced costs while maintaining broader hospital capacity.

Prepared by: Saung Hnin Phyu

AI Reflection

We asked ChatGPT for some help on how to calculate some parameters in our model, for refining sentence structure for the report (except for the discussion), and for debugging our code.

For the parameters, we asked ChatGPT how to calculate the recovery rate and infection risk. We wanted to use the recovery rate for continuous models, but did not know how to convert the rate for a discrete model.